🎓 Module 2 — AI Teacher (IFP Panel Only) Implementation & Technical Specification Report

Complete End-to-End Architectural Design, Requirements, Feature Specifications, 3rd-Party Cost Breakdown & Canvas Integration Roadmap

Target Platform: Android IFP Whiteboard Canvas | Architecture: Clean MVVM + Hilt DI + Jetpack Compose | AI Engine: Google Gemini 1.5 + ML Kit + Symja

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1. 🎓 Module 2 — AI Teacher (IFP Panel Only) Implementation Report

> 📌 **System Architecture Overview for IFP Panel Canvas**
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> Each AI feature is designed specifically for **IFP Panel Canvas Integration**, leveraging Bloom's infinite canvas (&mathbb;R^2 space), vector stroke engine, state management (`CanvasViewModel.kt`), and touch/stylus pipeline (`CanvasScreen.kt`).

🚀 Feature-by-Feature Implementation Specification

1. 📘 AI Lesson Planner (Interactive Curriculum Builder)

Feature Scope: Generates structured, multi-step lesson plans (Learning Objectives, Teaching Flow, Classroom Activities, Homework, and Key Outcomes) customized by Grade Level, Subject, and Duration.

- **Trigger:** Teacher opens IFP Top Bar → Taps **"AI Lesson Planner"**.
- **Input Panel:** Modal drawer for entering *Topic* (e.g. "Photosynthesis"), *Grade* (e.g. "Grade 8"), and *Duration* (e.g. "45 mins").
- **Canvas Output:** Generates an expanded **Lesson Board Frame** (`StrokeType.FRAME`) on the canvas containing 5 interactive cards: Objectives, Teaching Flow Timeline, Activities, Homework, and Expected Outcomes.

```
data class LessonPlanRequest(
    val topic: String,
    val gradeLevel: String,
    val durationMinutes: Int
)

data class LessonPlanResponse(
    val objectives: List<String>,
    val teachingFlow: List<FlowStep>,
    val activities: List<String>,
    val homework: List<String>,
    val outcomes: List<String>
)

// Canvas ViewModel Action: Places a framed structured layout on the infinite canvas
fun insertLessonPlanToCanvas(plan: LessonPlanResponse, origin: Offset) {
    val frameStroke = Stroke(
        strokeType = StrokeType.FRAME,
        points = listOf(origin, Offset(origin.x + 1200f, origin.y + 800f)),
        text = "Lesson Plan: ${plan.topic}"
    )
    // Add nested activity and flow cards inside the frame bounding box
}
```

2. 📊 AI PPT Generator (Editable Slide Deck & PPT/PDF Exporter)

Feature Scope: Generates editable slide presentations directly on the whiteboard canvas. Teachers can present slide-by-slide or edit text/diagrams using their stylus, then export the presentation as `.pptx` or `.pdf`.

- **Canvas Rendering:** Generates 16:9 aspect ratio **Slide Frames** laid out horizontally across the infinite canvas.
- **Presentation Mode:** Fullscreen presenter bar on the IFP panel with **Next / Previous Slide** controls and live ink annotation overlays.
- **Export Engine:** One-tap export to `.pptx` (via Apache POI) or `.pdf` (via Android `PdfDocument` rendering).

3. 💡 AI Explain (Tiered Conceptual Explainer)

Feature Scope: Provides explanations for any concept on the canvas at 3 complexity levels (**Beginner**, **Standard**, **Advanced**) using real-world examples, storytelling, or analogies.

- **Trigger:** Teacher circles or lasso-selects any text/ink on the board and selects **"AI Explain"**.
- **Canvas Placement:** Places an explanation callout card next to the selected concept.

4. 📝 AI Quiz Generator (Interactive On-Board Assessment)

Feature Scope: Generates customizable quizzes with multiple question formats (MCQ, True/False, Fill in the Blanks, HOTS [Higher Order Thinking Skills], Descriptive, and Difficulty Levels).

- **Teacher Reveal Button:** Tapping "Show Answer & Solution" on the IFP panel reveals answer keys with step-by-step reasoning.
- **Interactive Grading:** Students write answers with stylus on the IFP, and teacher taps "Grade Ink with AI" to evaluate responses live.

5. 📄 AI Worksheet Generator (Printable Homework & Practice Papers)

Feature Scope: Creates printable worksheets, practice papers, and homework sheets complete with student headers, instructions, marks distribution, and answer keys.

- **Canvas Layout:** Renders an A4-proportioned Canvas Frame (210mm × 297mm) complete with headers, Section A (MCQs), Section B (Short Answers), Section C (HOTS).
- **Print/Export:** One-tap print to Android network printer or export as PDF.

6. 🗺️ AI Mind Maps (Interactive Concept Maps, Flowcharts & Timelines)

Feature Scope: Generates vector mind maps, flow charts, historical timelines, and relationship maps from any topic or selected board notes.

- **Vector Synthesis:** Central Topic Node → Sub-topic Nodes → Vector Connecting Arrows.
- **Stylus Expansion:** Double-tapping any node on the IFP panel asks AI to expand that branch into deeper sub-nodes.

7. 🎨 AI Diagram Generator (Text to Labeled, Editable Diagrams)

Feature Scope: Converts text prompts (e.g. "Human Digestive System", "Solar System", "Electric Motor", "Plant Cell") into fully labeled, interactive, and editable diagrams directly on the canvas.

- **"Hide Labels" Quiz Mode:** Teacher taps "Hide Labels" for an interactive classroom quiz where students fill in labels using the stylus.

8. 🧮 AI Formula Solver (Math & Science Step-by-Step Solver with Graphing)

Feature Scope: Recognizes handwritten or typed equations ($\int (3x^2 + 2x)\, dx$, $E = mc^2$, $2x + 5 = 15$), solves them step-by-step, provides explanations, and plots 2D/3D Cartesian graphs on the canvas.

9. 📚 AI Summarizer (Revision Notes, Flashcards & Formula Sheets)

Feature Scope: Summarizes large sections of board notes into 4 structured review formats: **Revision Notes**, **Interactive Flash Cards**, **Formula Cheatsheets**, and **Exam Key Points**.

10. 🌐 AI Translator (10 Indian Languages)

Feature Scope: Translates handwritten or typed text on the IFP canvas into **10 Indian Languages**: **Hindi**, **Marathi**, **Gujarati**, **Tamil**, **Telugu**, **Kannada**, **Malayalam**, **Bengali**, **Punjabi**, and **Urdu**.

11. 🎙️ AI Voice Assistant (Hands-Free IFP Classroom Controller)

Feature Scope: Voice-activated assistant allowing teachers to command the IFP whiteboard hands-free while standing at the board (e.g., *"Create a 5-question quiz on Motion"*, *"Next slide"*, *"Explain this again simply"*, *"Generate homework sheet"*).

📊 Complete Summary Matrix for IFP Panel Canvas

#	AI Teacher Service	IFP Canvas Element	Output Type
1	**AI Lesson Planner**	Structured Frame (`StrokeType.FRAME`)	Multi-card teaching flow & objectives
2	**AI PPT Generator**	16:9 Slide Deck Frames	Presentation slides + PPTX/PDF Export
3	**AI Explain**	Context Callout Cards	3-tier explanation (Beginner/Std/Adv)
4	**AI Quiz Generator**	Interactive Quiz Cards	MCQs, T/F, HOTS + Answer Keys
5	**AI Worksheet Generator**	A4 Printable Frame	Standardized exam & practice papers
6	**AI Mind Maps**	Connected Vector Graph	Concept maps, flowcharts, timelines
7	**AI Diagram Generator**	Vector Graphic + Callout Labels	Labeled diagrams with Quiz Mode
8	**AI Formula Solver**	Math Step Card + Vector Plot	Step-by-step solution + 2D Cartesian Graph
9	**AI Summarizer**	Collapsible Revision Cards	Notes, Flash Cards, Formula Sheets
10	**AI Translator**	Bilingual Subtitle Text Blocks	10 Indian Languages Translation
11	**AI Voice Assistant**	Floating Audio Mic HUD	Voice-controlled IFP Canvas execution

3. 📋 Specification of AI Requirements

Here is the comprehensive specification of AI Requirements needed to power Module 2 — AI Teacher on the Whiteboard / IFP Panel:

3.1 AI Model & API Requirements

- **Primary Engine: Google Gemini API** (Gemini 1.5 Flash for low-latency speed; Gemini 1.5 Pro for complex diagrams & reasoning).
- **Capabilities:** Multimodal Vision Input, Strict JSON Output Schema Mode, System Prompt Role Instruction, Token Streaming (`Flow<String>`).
- **On-Device ML:** Google ML Kit Digital Ink Recognition API (handwriting OCR) + Google ML Kit Text Recognition v2.

3.2 Multimodal Data Ingestion Requirements

Input Type	Format	Source in App	AI Processing Goal
Stylus Ink Geometry	Raw Stroke Coordinates (x, y, t)	`Stroke.kt`	Recognize handwriting, formulas, & freehand sketches
Canvas Area Bounding Box	Rendered PNG Bitmap (300 DPI)	Screen Crop Bounding Box	Visual understanding of drawings, math, & diagrams
Reference Documents	Image (PNG/JPG) / PDF Bitmap	`Note.sourceUri`	Contextual Q&A, lesson planning, & summarization
Teacher Voice Audio	PCM 16kHz Audio Stream / Transcript	Android SpeechRecognizer / Mic HUD	Speech-to-Intent parsing for hands-free IFP control
Text & Markdown	Raw UTF-8 String	`Note.contentJson`	Translation, quiz generation, & mind mapping

3.3 Multilingual AI Requirements (10 Indian Languages)

The AI engine must support **10 Indian Languages** with accurate educational terminology: **Hindi**, **Marathi**, **Gujarati**, **Tamil**, **Telugu**, **Kannada**, **Malayalam**, **Bengali**, **Punjabi**, and **Urdu**, preserving

mathematical & scientific formulas during translation.

3.4 Performance & Latency Requirements

AI Task	Latency Target	Execution Target	Fallback Strategy
Handwriting OCR	< 150 ms	On-Device (ML Kit)	Cache locally
Voice Command Intent	< 500 ms	On-Device / Cloud Speech API	Show intent suggestion on screen
AI Explain / Formula Solve	< 1.5 seconds	Cloud (Gemini 1.5 Flash)	Local Symja engine for basic algebra
AI Quiz / Mind Map	< 2.5 seconds	Cloud (Gemini 1.5 Flash)	Show skeleton loader on canvas
AI PPT Deck / Diagram	< 4.0 seconds	Cloud (Gemini 1.5 Pro)	Stream slide-by-slide progress

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4. 💳 3rd-Party Accounts, API Keys, SDK Dependencies & Cost Breakdown

Here is the complete breakdown of all 3rd-Party Accounts, API Keys, SDK Dependencies, and Cost Estimations required to build and deploy Module 2 — AI Teacher:

4.1 Third-Party Accounts & Service Requirements

Service / API	Purpose	Account Required	API Key Required?
Google Gemini API	AI Teacher Engine (Lesson Plan, PPT, Quiz, Mind Maps, Diagrams, Formula, etc.)	**Google AI Studio / GCP Billing Account**	**Yes** (`GEMINI_API_KEY`)
Google ML Kit	Offline Handwriting OCR & Text Recognition	None (Bundled in Android App)	**No** (100% Local)
Android Speech Engine	Voice Assistant hands-free panel control	None (Built into Android OS)	**No** (Local OS Service)
Symja Math Engine	Step-by-step symbolic math & calculus evaluation	None (Open-Source Library)	**No** (Runs in App)
Apache POI	PPTX Slide Deck generation & export	None (Open-Source Library)	**No** (Runs in App)
Firebase (Optional)	Secure API key distribution & feature flags	**Firebase Console Account**	**Yes** (`google-services.json`)

4.2 Detailed API Pricing & Model Cost Breakdown

1. Gemini 1.5 Flash (90% of Features)

- 🆓 **Free Tier (Google AI Studio):** 15 RPM, 1M TPM, 1,500 RPD — **$0.00 / Month** (100% Free for Dev & Small Scale!).
- 💳 **Pay-As-You-Go Paid Tier (GCP Billing):**
  - Input Text: $0.075 per 1M Tokens.
  - Input Image: $0.00001875 per image (~53 images per 1 cent).
  - Output Text: $0.30 per 1M Tokens.

2. Gemini 1.5 Pro (10% of Features: Complex Labeled Diagrams & PPT Decks)

- 🆓 **Free Tier:** 2 RPM, 50 RPD — **$0.00 / Month**.
- 💳 **Pay-As-You-Go Paid Tier:** Input Text: $1.25 / 1M Tokens | Output Text: $5.00 / 1M Tokens.

3. On-Device & Open-Source Libraries

- **Google ML Kit Digital Ink Recognition:** **$0.00** (Free, runs on-device).

- **Android Native SpeechRecognizer:** $0.00 (Free).
- **Symja Math Engine & Apache POI:** $0.00 (Open Source Apache 2.0).

4.3 Total Monthly Cost Estimations by Scale

Deployment Scale	Daily Usage Volume	Total Monthly Cost	Per Panel Cost
Dev & Testing (1–3 IFP Panels)	~200 Queries/Day	**$0.00 / Month**	$0.00
Single School (10 IFP Panels)	~1,000 Queries/Day	**~$18.00 / Month**	~$1.80 / panel / month
School Network (100 IFP Panels)	~10,000 Queries/Day	**~$160.00 – $190.00 / Month**	~$1.60 – $1.90 / panel / month

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5. 📄 Canvas AI Services Integration Report & Architectural Roadmap

> 🗺️ **Executive Integration Blueprint**
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> This section details the step-by-step technical plan to integrate all 11 AI Teacher services directly into the existing whiteboard canvas codebase (`CanvasScreen.kt` , `CanvasViewModel.kt` , `Stroke.kt` , `Note.kt` , `CanvasTransform.kt` , Hilt DI modules) when ready for development execution.

5.1 Step-by-Step Canvas Implementation Strategy

Step 1: Data Contracts & Domain Models (`domain/model/`)

Create domain entities for structured AI responses:

- `LessonPlan` : Topic, objectives, flow steps, activities, homework.
- `PresentationDeck` : List of 16:9 slide items (title, bullet points, layout).
- `QuizQuestion` : Question text, options, answer key, solution, question type.
- `MindMapGraph` : Root node with radial children and vector arrow connectors.

- `LabeledDiagram`: Vector paths, title, label anchor points (x_i, y_i).
- `FormulaSolution`: Latex string, step-by-step algebra list, function curve y = f(x).

Step 2: AI Service & Repository Layer (`data/repository/` & `di/`)

- Implement `AiTeacherRepositoryImpl` using the Google Generative AI SDK (`GenerativeModel`) for Gemini 1.5.
- Enforce **Strict JSON Output Schemas** so Gemini returns validated JSON for canvas entities.
- Inject `AiTeacherRepository` via Hilt DI in `AiTeacherModule.kt`.

Step 3: State Integration in `CanvasViewModel.kt`

Extend `CanvasState` with AI state flags:

```
data class CanvasState(
    // Existing fields: strokes, scale, offsetX, offsetY, selectedStrokeIds...
    val isAiLoading: Boolean = false,
    val aiProgressText: String = "",
    val activeAiModal: AiTeacherModalType? = null,
    val isVoiceAssistantListening: Boolean = false,
    val activePresentationMode: Boolean = false,
    val currentSlideIndex: Int = 0
)
```

Step 4: World Space Placement Math (`CanvasTransform.kt`)

When AI generates an output (e.g. a Lesson Plan, Mind Map, or Diagram), it is placed directly in front of the teacher's current viewport in **World Space Coordinates (x_w, y_w)**:

$Origin_w$ = (-offsetX / scale + 50dp, -offsetY / scale + 50dp)

5.2 Detailed How-To Plan for Each of the 11 AI Services

1. 📘 AI Lesson Planner on Canvas

1. Teacher taps **"AI Lesson Planner"** on the IFP dock.
2. Prompt modal collects Topic, Grade Level, and Duration.
3. `CanvasViewModel` calls Gemini API and receives structured JSON.
4. Generates a master `StrokeType.FRAME` titled `"Lesson Plan: [Topic]"` with 5 nested card objects arranged vertically.
5. Teacher can drag, edit, or reorder any card using the stylus.

2. 📊 AI PPT Generator & Slide Deck Engine

1. Teacher requests a 5-slide deck on a topic.

2. Generates 5 horizontal 16:9 Slide Frames (`StrokeType.FRAME`) on the canvas separated by 40dp gaps.
3. **Presentation Mode:** Fullscreen toolbar appears with Next Slide / Prev Slide buttons, smoothly panning canvas viewport from slide to slide.
4. **Export:** One-click export button calls `PptxExporter` (Apache POI) to generate `.pptx` or `.pdf`.

3. 💡 AI Explain (3-Tier Conceptual Explainer)

1. Teacher uses Lasso to select any handwritten text or drawing on the board.
2. Floating Lasso bar shows **"AI Explain"** → Selects level (Beginner, Standard, Advanced) and style (Analogy, Story, Real World).
3. Renders a callout card alongside the selected concept with styled markdown text.

4. 📝 AI Quiz Generator & Interactive Grading

1. Teacher generates a quiz (MCQ, True/False, Fill in Blanks, HOTS).
2. Renders interactive question cards on the canvas.
3. **Show/Hide Answer:** Teacher taps a button on the card to reveal answer keys.
4. **AI Ink Grading:** Students write answers on the IFP with stylus; teacher taps "Grade Ink with AI", which crops handwritten answer, runs OCR, and displays evaluation feedback.

5. 📄 AI Worksheet Generator (Printable A4 Layout)

1. Teacher requests a homework sheet or practice paper.
2. Creates an A4-proportioned Canvas Frame (210mm × 297mm) containing formatted headers, mark distributions, and questions.
3. Integrated with Android `PrintManager` to print directly to a classroom network printer.

6. 🗺️ AI Mind Maps & Vector Concept Trees

1. Teacher enters a topic (e.g. "Photosynthesis").
2. Radial Layout Algorithm calculates node coordinates in World Space:
   $x_n = x_{center} + R \cdot \cos(\theta_n),\ y_n = y_{center} + R \cdot \sin(\theta_n)$
3. Renders central topic box, sub-topic sticky notes, and connecting arrow strokes (`StrokeType.ARROW`).
4. Double-tapping any node requests Gemini to expand that specific branch.

7. 🎨 AI Diagram Generator with "Hide Labels" Quiz Mode

1. Teacher requests a diagram (e.g., "Human Digestive System").
2. Generates vector paths / illustration + callout line pointers + text label blocks.
3. **"Hide Labels" Mode:** Toggling "Hide Labels" hides text labels, leaving blank box pointers so students write labels on the IFP screen with stylus.

8. 🧮 AI Formula Solver & 2D Function Plotter

1. Teacher circles a handwritten math equation ($\int x^2 dx$ or $y = x^2 - 4$).
2. ML Kit / Gemini parses formula → Symja evaluates step-by-step algebra.
3. Plots 2D Cartesian function graph curve directly on canvas using `Canvas.drawPath()`.

9. 📚 AI Summarizer

1. Teacher selects a region of board notes → Taps **"AI Summarizer"**.
2. Generates 4 tabbed canvas cards: Revision Notes, Interactive Flash Cards, Formula Sheet, Exam Key Points.

10. 🌐 AI Translator (10 Indian Languages)

1. Teacher selects canvas text → Chooses target language (Hindi, Marathi, Gujarati, Tamil, Telugu, Kannada, Malayalam, Bengali, Punjabi, Urdu).
2. Performs in-place text translation or renders bilingual subtitle blocks underneath.

11. 🎙️ AI Voice Assistant (Hands-Free IFP Controller)

1. Microphone HUD stays docked in bottom corner of IFP panel.
2. Listens for voice triggers (*"Create a 5-question quiz on Motion"*, *"Next slide"*, *"Explain again"*, *"Translate to Hindi"*).
3. Speech-to-Intent parser directly triggers `CanvasViewModel` methods hands-free.

5.3 Fallback & Offline Operational Strategy

- **Online Mode:** Uses Google Gemini 1.5 Flash / Pro API for full multimodal reasoning.
- **Offline Mode:** Automatically falls back to **Google ML Kit** for local handwriting OCR and **Symja Engine** for math solving when disconnected.
- **Non-Destructive Canvas History:** All AI-placed canvas elements are recorded in the `CanvasViewModel` Undo/Redo stack, allowing teachers to revert any AI action with 1 tap.